

1 Production order data SAP system → HYDRA

Data to the production order are generally transferred completely with all corresponding information (series of operations, components etc.) and thus in a self-contained way to HYDRA (download).

When transferring a production order to HYDRA this order could – according to this – already be available and have been transferred because of a modification or a new creation is concerned. The procedure of the transfer to HYDRA is described in detail in the following chapters.

Creating/modifying data

All data that are relevant for HYDRA are completely summarized to a IDOC and transferred to HYDRA via the message function (= file extension) “APP” at the interface for the transfer of production orders.

Such an IDOC

- contains one production orders with
- segments which are divided hierarchically and contain the single data.

When transferring the order data from the SAP system to HYDRA it should be taken into consideration that these data are transmitted to HYDRA in a specific sequence in case of an initial data transfer. This means that the order data described in the following chapters have to be transferred to HYDRA in the sequence/hierarchy that is given there. The sequence can be chosen arbitrarily within one transfer stage, for example the series of operations 0200 can be transmitted prior to the series of operations 0100.

Modifications to series of operations having the status “running”, “finished” or “deleted” are not allowed in the standard of HYDRA.

Generally, HYDRA always expects a coherent data structure, which means that unused data fields have to be assigned to default values specified according to the convention.

Deleting data

A deletion download deletes those operations that are no longer necessary in the production process from the HYDRA database.

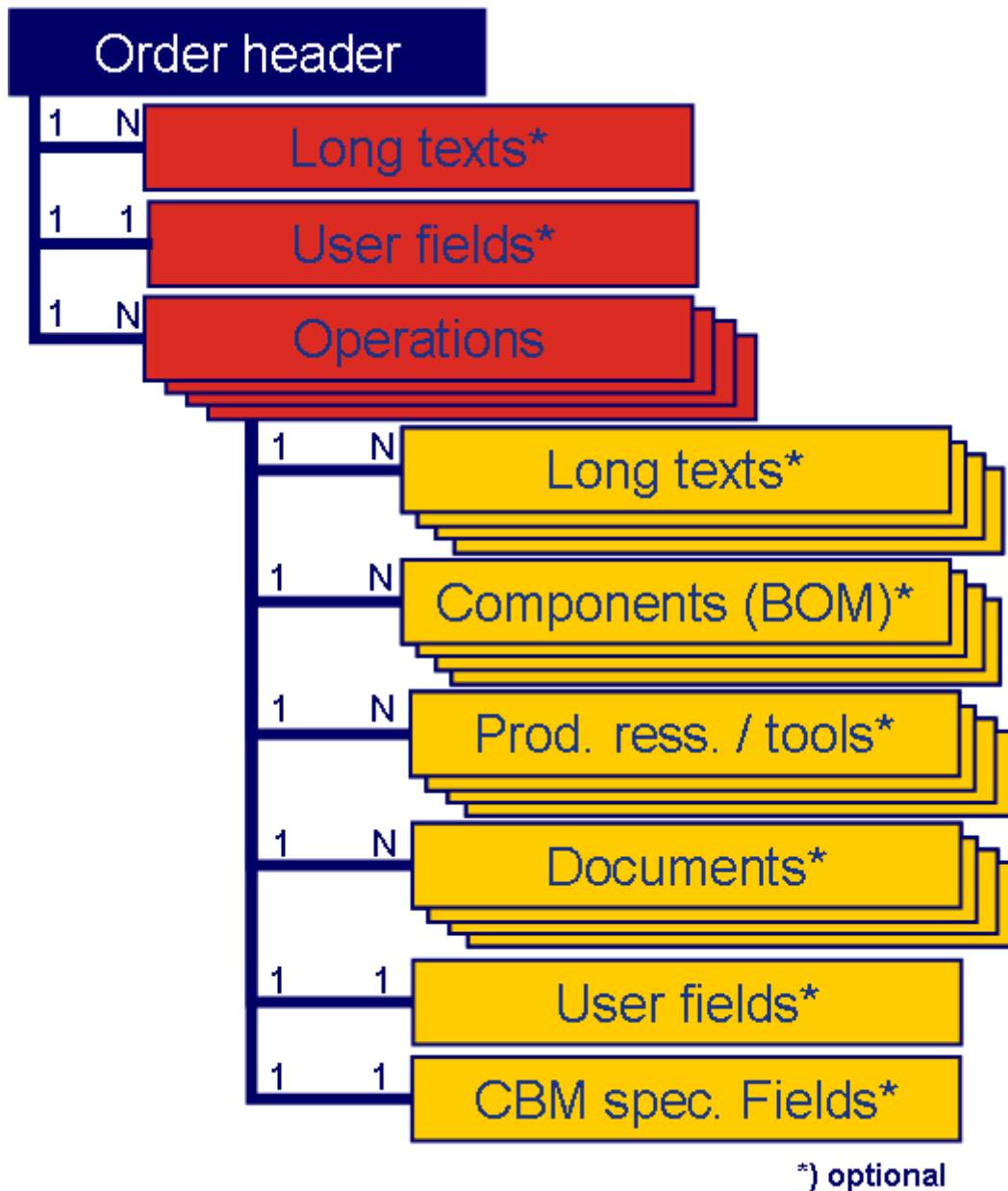
As with the modifications, the deletion result depends on the operation's status. Operations identified as "Running", "Interrupted", "Finished" or "Deleted", will not be deleted.

Moreover, the confirmation number (CONF_NO) will be checked to identify the orders via order, sequence and transaction. This prevents any accidental deletion of orders that are seemingly the same but which have different confirmation numbers.

Order and operation data is deleted using the E2BP_PP_PDC_OPERA1000 segment that is derived from SAP standard interface PP-PDC. The segments contains the keys of the operation to be deleted.

IDOC structure

Order-related data are transferred per production order in a multi-stage IDOC which has the following structure:



The following specifications result from this for the IDOC:

Message type:	ZPPORDER
Idoc type:	ZPPORDER02
Message functions:	APP (creating/deleting data) DEL (deleting data)
Segments:	Z2AU_HD000X000 (order header) - Z2AU_INFO_AI000X000 (long texts) - Z2AU_USRFLD000X000 (user fields) - Z2AG_HD000X000 (data of series of operations) - Z2AG_KOMPL000X000 (Components list) - Z2AG_KOMPL_USRFLD000X000 - Z2AG_FHM000X000 (production resources and tools/resources) - Z2AG_DOC000X000 (documents) - Z2AG_INFO_AI000X000 (long texts) - Z2AG_USRFLD000X000 (user fields) - Z2AG_RF000X000 (MPL-RF-specific data) - Z2AG_ANETZ000X000 (order network) - Z2MD_PRODVAR000X000 (production variants) - E2BP_PP_PDC1000 (delete operations)



For creation of the segments for HYDRA inbound, the needed segments in the ALE-development (in SAP) must be defined in the scheme: Z1<segment name>. When versioning in SAP the segments are created in the form: Z2<segment name><version>.

Example: **Z1AU_HD000X** will result in **Z2AU_HD000X000**

The structures belonging to the single segments are described as follows. The single columns have the following meaning:

Column	Meaning
Field	Designation of the field
V(use)	S It is a key field which identifies the data record clearly if needed together with other fields characterized as key fields. The field must be filled. M It is a mandatory field which has to be filled. ML Mandatory field if the control center of HYDRA is used (HLS). MM Mandatory field if the material and production logistics of HYDRA is used (MPL or MPL/RF). K Field is allowed to stay empty.
T(type)	Data type according to the description in the chapter above.
L(length)	Field length for fields of the data type DEC: total number of places, without decimal point and algebraic sign.
D(decimal places)	For fields of the data type DEC: number of places after decimal point, otherwise: not relevant
Description	Description of the field or comment to the field

Header (Z2AU_HD000X000)

Field	V	T	L	D	Description
AUNR	S	CHAR	12		Order number
AUART	M	CHAR	5		Order type Fix „0“
ATK	M	CHAR	40		Producing material (Article); alphabetical characters in the UPPER CASE
ATKBEZ	K	CHAR	40		Designation of the article
KDBEZ	K	CHAR	40		Designation of the customer
KDAUNR	K	CHAR	25		Customer order
KDAUPOS	K	CHAR	15		Customer order position
EXTPRIO	M	CHAR	1		Priority
AUIDX	K	DEC	5	2	Order index; should be 0.00. Information: According to the description of the data type it should be considered that an algebraic sign and a decimal point have to be supplemented to the length that is indicated here. This is valid for all following fields of the data type DEC.
SGE_B	M	CHAR	3		Base quantity unit
SGR_GUTB	M	DEC	13	3	Target quantity (base quantity unit)
SGR_AUSB	K	DEC	13	3	Target scrap (base quantity unit)
MATTYP	M	CHAR	20		Material type (configured in HYDRA) of the producing material (article)
CNR	K	CHAR	20		Batch number, no processing in HYDRA
PCNR	K	CHAR	20		Inspection order/inspection lot number
PPKTYP	K	CHAR	1		Physical-sample category
DATFB	ML	DATE	8		Earliest start (date) → SAP: Start BasicDates
ZEIFB	ML	TIME	6		Earliest start (time) → SAP: Start BasicDates () Please note: 240000 is not allowed; use instead 235959
DATSE	ML	DATE	8		Latest end (date) → SAP: Finish BasicDates
ZEISE	ML	TIME	6		Latest end (time) → SAP: Finish BasicDates Please note: 240000 is not allowed; use instead 235959
DATTERM	K	DATE	8		Scheduled start (date)
ZEITERM	K	TIME	6		Scheduled start (time) Please note: 240000 is not allowed; use instead 235959
DATERME	K	DATE	8		Scheduled end (date)
ZEITERME	K	TIME	6		Scheduled (time) Please note: 240000 is not allowed; use instead 235959
TERMART	K	CHAR	1		Scheduling type. Mandatory field if the scheduling is supposed to take place in HYDRA. V = Forward R = Backward
REDSTRAT	K	CHAR	2		→ Reduction strategy. According to HYDRA Customizing
AUGRP	K	CHAR	4		Order group belongs to "production scheduler" in SAP
DISP	K	CHAR	10		MRP controller
PRJNR	K	CHAR	25		Project number; belongs to the WBS number in SAP
PLANAUNR	K	CHAR	25		Planned order

Field	V	T	L	D	Description
KTR	K	CHAR	25		Cost object
APNR	K	CHAR	40		Working plan
APVER	K	CHAR	12		Working plan planned order version
SLVER	K	CHAR	12		Bill of material version
KLKK_MNR	K	DEC	13	3	Calculated costs machine
KLKK_L	K	DEC	13	3	Calculated costs wage
KLKK_MAT	K	DEC	13	3	Calculated costs material
KLKK:SONST	K	DEC	13	3	Calculated costs other
MATWERT_G UT	K	DEC	13	3	Material value
MATWERT_A US	K	DEC	13	3	Scrap value
KBN:LBEZID	K	CHAR	15		eKANBAN logical system
ATKIDX	K	CHAR	50		Drawing issue number

Long texts of the order (Z2AU_INFO_AI000X000)

Text fields that are displayed at the HYDRA console can be transferred to HYDRA as additional information to the order (header) by means of the following structure. Each data record contains one page with max. 10 lines and 80 characters text information each.

Information

This data structure is only available in connection with the license HYD-INF or HKMPP-INF.

Field	V	T	L	D	Description
KEY	S	CHAR	12		Order number
TYP	S	CHAR	2		Record type; fix: "AI"
SUBKEY_1	S	NUM	8		Reserved; assign fix to "00000000"
SUBKEY_2	S	NUM	8		Consecutive numbering starting with "00000001" within the key
INFO_BEZ	K	CHAR	20		Short text. If it is left empty the first 20 lines of info text 1 are adopted
INFO_1	K	CHAR	80		Info text 1
INFO_2	K	CHAR	80		Info text 2
INFO_3	K	CHAR	80		Info text 3
INFO_4	K	CHAR	80		Info text 4
INFO_5	K	CHAR	80		Info text 5
INFO_6	K	CHAR	80		Info text 6
INFO_7	K	CHAR	80		Info text 7
INFO_8	K	CHAR	80		Info text 8
INFO_9	K	CHAR	80		Info text 9
INFO_10	K	CHAR	80		Info text 10

User fields of the order header (Z2AU_USRFLD000X000)

User fields offer the possibility to file further customer-specific information in HYDRA besides the fields that are available in the HYDRA standard. The segment Z2AU_USRFLD000X000 serves for the transfer of these data from the SAP system to HYDRA and for assigning them to the order (header).

What user fields are concerned here and which meaning they have is determined via the so-called user field key. Each user field key describes a combination of user fields. The customer-specific configuration of the user field key is carried out in the scope of the HYDRA customizing.

Information

When it comes to the data exchange of user fields between the SAP system and HYDRA the customer has to guarantee that the user field keys are maintained equally in both systems so that a consistent data exchange is possible between both systems.

This data structure is only available in connection with the license HYD-INF or HKMPP-INF.

Field	V	T	L	D	Description
AUNR	S	CHAR	12		Order number
USRFLD	S	CHAR	8		User field key
FU_1	K	DATE	8		User field 1
FU_2	K	DATE	8		User field 2
FU_3	K	DATE	8		User field 3
FU_4	K	DATE	8		User field 4
FU_5	K	DATE	8		User field 5
FU_6	K	DATE	8		User field 6
FU_7	K	NUM	8		User field 7
FU_8	K	NUM	8		User field 8
FU_9	K	NUM	8		User field 9
FU_10	K	NUM	8		User field 10
FU_11	K	NUM	8		User field 11
FU_12	K	NUM	8		User field 12
FU_13	K	NUM	8		User field 13
FU_14	K	NUM	8		User field 14
FU_15	K	NUM	8		User field 15
FU_16	K	NUM	8		User field 16
FU_17	K	NUM	8		User field 17
FU_18	K	NUM	8		User field 18
FU_19	K	NUM	8		User field 19
FU_20	K	NUM	8		User field 20
FU_21	K	NUM	8		User field 21
FU_22	K	NUM	8		User field 22
FU_23	K	DEC	13	3	User field 23
FU_24	K	DEC	13	3	User field 24
FU_25	K	DEC	13	3	User field 25

Field	V	T	L	D	Description
FU_26	K	DEC	13	3	User field 26
FU_27	K	DEC	13	3	User field 27
FU_28	K	DEC	13	3	User field 28
FU_29	K	CHAR	1		User field 29
FU_30	K	CHAR	1		User field 30
FU_31	K	CHAR	1		User field 31
FU_32	K	CHAR	1		User field 32
FU_33	K	CHAR	1		User field 33
FU_34	K	CHAR	1		User field 34
FU_35	K	CHAR	1		User field 35
FU_36	K	CHAR	1		User field 36
FU_37	K	CHAR	1		User field 37
FU_38	K	CHAR	1		User field 38
FU_39	K	CHAR	1		User field 39
FU_40	K	CHAR	1		User field 40
FU_41	K	CHAR	1		User field 41
FU_42	K	CHAR	1		User field 42
FU_43	K	CHAR	1		User field 43
FU_44	K	CHAR	1		User field 44
FU_45	K	CHAR	10		User field 45
FU_46	K	CHAR	10		User field 46
FU_47	K	CHAR	10		User field 47
FU_48	K	CHAR	10		User field 48
FU_49	K	CHAR	10		User field 49
FU_50	K	CHAR	10		User field 50
FU_51	K	CHAR	20		User field 51
FU_52	K	CHAR	20		User field 52
FU_53	K	CHAR	20		User field 53
FU_54	K	CHAR	20		User field 54
FU_55	K	CHAR	20		User field 55
FU_56	K	CHAR	20		User field 56
FU_57	K	CHAR	20		User field 57
FU_58	K	CHAR	20		User field 58
FU_59	K	CHAR	20		User field 59
FU_60	K	CHAR	20		User field 60
FU_61	K	CHAR	20		User field 61
FU_62	K	CHAR	20		User field 62
FU_63	K	CHAR	20		User field 63
FU_64	K	CHAR	20		User field 64
FU_65	K	CHAR	40		User field 65
FU_66	K	CHAR	40		User field 66

Operation (Z2AG_HD000X000)

The operation-related production specifications and data are contained in the segment Z2AG_HD000X000.

Field	V	T	L	D	Description																		
AUNR	S	CHAR	12		(Production-)Order number																		
AFOLG	S	NUMC	6		Sequence																		
AGNR	S	CHAR	4		Operation																		
UVGNR	S	CHAR	4		Sub-operation (not supported)																		
AART	S	CHAR	5		Order type fix "0" production order																		
AGBEZ	M	CHAR	40		Designation of the operation																		
ATK	K	CHAR	40		Number of producing material; Article (number) Alphabetical characters in UPPER CASE. Remark: in HYDRA only 25 digits are displayed.																		
ATKBEZ	K	CHAR	40		Designation/ Description of the material/article																		
MATTYP	MM	CHAR	20		Material type of the article; if HYDRA MPL is used according to the configuration of the material type																		
EXTPRIO	M	CHAR	1		Priority (0 - 9; 9 = high priority)																		
MNR	M	CHAR	8		Planned workplace At least one of the fields MNR and/or MGRP must be transferred. If the workplace is transferred, HYDRA will determine the workplace's group according to the configuration in order to avoid inconsistent data. This means that the transferred group will be ignored.																		
MGRP	M	CHAR	8		Planned group and/or group of the workplace At least one of the fields MNR and/or MGRP must be transferred. If no machine group is transferred, the system will determine automatically the machine group using the transferred workplace.																		
OPT_PLAN	M	CHAR	1		Planned M - operation is planned (in detail) to the workplace G - operation is in the pool of (groups) (MNR empty) Depending on the workplace (MNR field) - the indicator will be set internally according to the following logic: <table><tr><td>Workplace</td><td>Planned</td><td>Result</td></tr><tr><td><u>MNR</u></td><td><u>OPT:PLAN</u></td><td></td></tr><tr><td>blank</td><td>blank/ "M"/"G"</td><td>OPT:PLAN will be set to "G"</td></tr><tr><td>not blank</td><td>blank</td><td>OPT:PLAN will be set to "M"</td></tr><tr><td>not blank</td><td>"M"</td><td>OPT:PLAN will stay on "M"</td></tr><tr><td>not blank</td><td>"G"</td><td>OPT:PLAN will stay on "G"</td></tr></table>	Workplace	Planned	Result	<u>MNR</u>	<u>OPT:PLAN</u>		blank	blank/ "M"/"G"	OPT:PLAN will be set to "G"	not blank	blank	OPT:PLAN will be set to "M"	not blank	"M"	OPT:PLAN will stay on "M"	not blank	"G"	OPT:PLAN will stay on "G"
Workplace	Planned	Result																					
<u>MNR</u>	<u>OPT:PLAN</u>																						
blank	blank/ "M"/"G"	OPT:PLAN will be set to "G"																					
not blank	blank	OPT:PLAN will be set to "M"																					
not blank	"M"	OPT:PLAN will stay on "M"																					
not blank	"G"	OPT:PLAN will stay on "G"																					
RES_WNR	K	CHAR	40		(Main) tool; is displayed in versatile evaluations; if empty it is filled during the FHM takeover (please see below).																		
RES_DNC	K	CHAR	40		NC-program; is displayed in versatile evaluations; if empty it if filled during the FHM takeover (please see below).																		
RES_EMAT	K	CHAR	40		(Main) input material; is displayed in versatile evaluations; if empty it is filled during the takeover of the components (please see below).																		
COLOR	K	CHAR	20		Color of the material																		
KST	K	CHAR	8		Cost center; no processing in HYDRA (only information)																		
KART	K	CHAR	10		Cost type; no processing in HYDRA (only information)																		
ASTUFE	K	CHAR	1		Authorization level for logging on/logging off OPs (lowest authorization = 1)																		
RMNR	K	CHAR	10		Confirmation number																		
DATTERM B	K	DATE	8		Scheduled start (date)																		
ZEITERMB	K	TIME	6		Scheduled start (time) Please note: 240000 is not allowed; use instead 235959																		
DATTERM E	K	DATE	8		Scheduled end (date)																		

Field	V	T	L	D	Description
ZEITERME	K	TIME	6		Scheduled end (time) Please note: 240000 is not allowed; use instead 235959
DATFB	K	DATE	8		Earliest start (date)
ZEIFB	K	TIME	6		Earliest start (time) Please note: 240000 is not allowed; use instead 235959
DATSB	K	DATE	8		Latest start (date)
ZEISB	K	TIME	6		Latest start (time) Please note: 240000 is not allowed; use instead 235959
DATFE	K	DATE	8		Earliest end (date)
ZEIFE	K	TIME	6		Earliest end (time) Please note: 240000 is not allowed; use instead 235959
DATSE	K	DATE	8		Latest end (date)
ZEISE	K	TIME	6		Latest end (time)
DATB	K	DATE	8		Planned start (date)
ZEIB	K	TIME	6		Planned start (time) Please note: 240000 is not allowed; use instead 235959
DATE	K	DATE	8		Planned end (date)
ZEIE	K	TIME	6		Planned end (time) Please note: 240000 is not allowed; use instead 235959
SGR_GUTB	K	DEC	13	3	Target quantity (base quantity unit)
SGR_GUTP	K	DEC	13	3	Target quantity (primary quantity unit)
SGR_GUTS	K	DEC	13	3	Target quantity (secondary quantity unit)
SGR_GUTT	K	DEC	13	3	Target quantity (tertiary quantity unit)
SGR_AUSB	K	DEC	13	3	Target scrap (base quantity unit)
SGR_AUSP	K	DEC	13	3	Target scrap (primary quantity unit)
SGR_AUSS	K	DEC	13	3	Target scrap (secondary quantity unit)
SGR_AUST	K	DEC	13	3	Target scrap (tertiary quantity unit)
SGE_B	K	CHAR	3		Base quantity unit
SGE_P	K	CHAR	3		Primary quantity unit of entry
SGE_S	K	CHAR	3		Secondary quantity unit of entry
SGE_T	K	CHAR	3		Tertiary quantity unit of entry
WEIGMENGE	K	DEC	13	3	Minimum send-ahead quantity (primary quantity unit)
MENGEPROZ_UNTLI	K	DEC	13	3	Underdelivery in per cent

Field	V	T	L	D	Description
OPT_UNTLI	K	CHAR	1		Reaction to underdelivery
MENGEPROZ_UEBLI	K	DEC	13	3	Overdelivery in per cent
OPT_UEBLI	K	CHAR	1		Reaction to overdelivery
UMRFAKTP_Z	K	NUM	8		Factor for conversion primary quantity → base quantity (counter)
UMRFAKTP_N	K	NUM	8		Factor for conversion primary quantity → base quantity (denominator)
UMRFAKTS_Z	K	NUM	8		Factor for conversion secondary quantity → base quantity (counter)
UMRFAKTS_N	K	NUM	8		Factor for conversion secondary quantity → base quantity (denominator)
UMRFAKTT_Z	K	NUM	8		Factor for conversion tertiary quantity → base quantity (counter)
UMRFAKTT_N	K	NUM	8		Factor for conversion tertiary quantity → base quantity (denominator)
RUEZ	K	NUM	8		Setup time in seconds
RUEZ_ZUSCHL	K	NUM	8		Addition of setup time in seconds
BEARBZEI	K	NUM	8		Processing time in seconds
PZ	K	NUM	8		Inspection time in seconds
ABRZ	K	NUM	8		Teardown time in seconds
LIZ	K	NUM	8		Delivery time in seconds
FREMDFERT	K	CHAR	1		External processing OP J/N
RLZ_EXPR	K	CHAR	6		Formula pro calculating the estimated remaining processing time (formula 1) Mainly relevant if the control center of HYDRA is used (HLS). Deviating settings are possible within the scope of the customizing of HYDRA.
RLZ_EXPR2	K	CHAR	6		Remaining running time (formula 2); optional (leave empty)
VLZ	K	NUM	8		Lead time in seconds
LIEZ_MAX	K	NUM	8		Max. synchronization time in seconds
WARTZ	K	NUM	8		Wait time in seconds
WARTZ_MIN	K	NUM	8		Minimal waiting time in seconds
LIEZ	K	NUM	8		Idle period in seconds
LART	K	CHAR	4		Wage type
AKKORD	K	CHAR	1		Piece rate indicator/piece-rate premium
TE	K	DEC	13	3	Premium specification te in seconds per 1000 pieces
TR	K	DEC	13	3	Premium specification tr in seconds
TEB	K	DEC	13	3	Premium specification teb in seconds per 1000 pieces
TRB	K	DEC	13	3	Premium specification trb in seconds
VERARBCODE	M	CHAR	6		Processing code; fix "SYSTEM" Deviating settings are possible within the scope of the customizing of HYDRA.
OPT_ERF	M	CHAR	1		Can be entered J/N
OPT_MULTIMNR	M	CHAR	1		Operation can be logged on parallel on different workplaces
OPT_CNR	MM	CHAR	1		Subject to batch tracing J/N
OPT_SNR	M	CHAR	1		Subject to management in serial numbers J/N ("J" only relevant for ADE-SNR)
SZY	K	NUM	8		Target cycle in seconds/1000, should be set; mandatory for MDE monitoring of the cycle of the machine
TLG	K	NUM	8		Partitioning; should be pre-defined with 1; mandatory for MDE monitoring of the cycle of the machine
IMPFAKT	K	DEC	13	3	Pulse factor; reserved; should be pre-defined with 1
OPT_SPLIT	K	CHAR	1		Can be split "V" (YES)/ "N" (NO)
MAXANZSPLIT	K	NUM	8		Max. number of splits (only relevant if OPT:SPLIT = "V")

Field	V	T	L	D	Description
MBVERH_RUE	K	DEC	5	2	Relation machines-user setting up; reserved.
MBVERH_NORM	K	DEC	5	2	Relation machines-user producing;
QUAL:NORM	K	NUM	8		PEP: Qualifikation Fertigen
QUAL:RUE	K	NUM	8		PEP: Qualifikation Rüsten
ATKIDX	K	CHAR	50		Drawing issue number
WERK_S	M	CHAR	4		SAP Plant

Long texts of the series of operations (Z2AG_INFO_AI000X000)

Text fields that are displayed at the HYDRA console can be transferred to HYDRA as additional function to the series of operations by means of the following structure. Each data record contains one page with max. 10 lines and 80 characters each.

It's displayed at the HYDRA client and at the terminal software.

Information

This data structure is only available in connection with HYD-INF or HKMPP-INF.

Field	V	T	L	D	Description
AUNR	S	CHAR	12		(Production-)Order number
AFOLG	S	NUMC	6		Sequence
AGNR	S	CHAR	4		Operation
UVGNR	S	CHAR	4		Sub-operation
TYP	S	CHAR	2		Record type; fix: "AI"
SUBKEY_1	S	NUM	8		Reserved; fix "00000000"
SUBKEY_2	M	NUM	8		Consecutive numbering starting with 1 within the key; if more than 10 lines are needed then the next record of this operation has the number 2
INFO_BEZ	K	CHAR	20		Short text; only relevant for SUBKEY:2 = "00000001". If empty the first 20 digits of info text 1 are adopted.
INFO_1	K	CHAR	80		Info text 1
INFO_2	K	CHAR	80		Info text 2
INFO_3	K	CHAR	80		Info text 3
INFO_4	K	CHAR	80		Info text 4
INFO_5	K	CHAR	80		Info text 5
INFO_6	K	CHAR	80		Info text 6
INFO_7	K	CHAR	80		Info text 7
INFO_8	K	CHAR	80		Info text 8
INFO_9	K	CHAR	80		Info text 9
INFO_10	K	CHAR	80		Info text 10

Components (Z2AG_KOMPL000X000)

Note

For order combination (cutting plan - extrusion/ slitting department) the component list has to be transferred ONLY for the mother operation!!

Field	V	T	L	D	Description
AUNR	S	CHAR	12		(Production-)Order number
AFOLG	S	NUMC	6		Sequence
AGNR	S	CHAR	4		Operation
UVGNR	S	CHAR	4		Sub-operation
ATK	S	CHAR	40		Material number; alphabetical characters in UPPER CASE
ATKBEZ	M	CHAR	40		Material designation
BEZ	K	CHAR	30		Comment 1
BEZ_2	K	CHAR	30		Comment 2
SLP	S	CHAR	10		Bill of material item / Position MPL-RF: Should position of the component in the layer structure (lamination) If several components are used in one operation, each component must have a unique BOM item. It is not allowed for two components to have the same BOM item.
SLS	M	NUM	8		Level of bill of material Material components with level of bill of material > 1 are always saved in the material type "I" = info component
ART	MM	CHAR	2		Material nature "M" (Consumable) material "T" Carrier material (only MPL-RF) "I" Info component
MATTYP	MM	CHAR	10		MPL/MPL-RF: material type If HYDRA MPL is used a valid material type that is configured in HYDRA has to be indicated
VERBR	MM	CHAR	1		MPL/MPL-RF: Consumption type Unless otherwise noted this field has to pre-assigned to "L".
OPT_ERSB	MM	CHAR	1		MPL-RF: replaceable – may another material than the planned one be used for such a component? Only the same material type as the one of the material to be produced may be used then. Unplanned material could only log on at machine if correct machine status is set. MPL/MPL-RF: J/N Otherwise: Fix "N"
OPT_WZW	MM	CHAR	1		MPL/MPL-RF: subject to changes; input batch change for a batch of this material forces an output batch change: if ART = "T" or "Z" → OPT:WZW must be "J" if ART = "I" or "A" → OPT:WZW must be "N" if ART = "M" → OPT:WZW: "J" or "N"
SGR_GUT	MM	DEC	13	3	MPL/MPL-RF: Required quantity for this component (position) in relation to the production of 1 unit/article in the primary quantity unit of the operation
SGE_GUT	MM	CHAR	3		MPL/MPL-RF: quantity unit of the required quantity
MENGEPROZ	K	DEC	13	3	Required quantity in per cent Reserved, currently no use
OTG	K	DEC	13	3	Upper tolerance limit in per cent; 3 places after decimal point Reserved; currently no use
UTG	K	DEC	13	3	Lower tolerance limit in per cent; 3 places after decimal point Reserved; currently no use

Field	V	T	L	D	Description
EGR_GUT	K	DEC	13	3	Total quantity/requirement quantity: total quantity required for OP, i.e. for the quantity to be produced (output quantity)
EGE_GUT	K	CHAR	3		Unit of the quantity required
SLS_M	K	NUMC	8		Level of the bill of material of the mother material
SLP_M	K	CHAR	10		Position of the bill of material of the mother material
MENGE_FIX	K	CHAR	1		Indicator: fixed quantity
PPS_RETRO	K	CHAR	1		Indicator: Backflush (in the SAP)
OPT_SLOS	K	CHAR	1		Flag „Collective Batch“ Available with MPL82 / MLE-Variant ZPPORDER_SAP_008 Indicating the type of component when working with serial numbers for the union of serial numbers: „ “ In case no serial numbers are used or, when serial numbers are used, to indicate serial number managed components, which ID is not forwarded to the next process step „F“ In case serial numbers are used to indicate the component, from which the ID is taken into the next processing step
VERB_ZAEHLER	K	CHAR	1		Flag Automatic counter consumption Available with MPL82 / MLE-Variant ZPPORDER_SAP_008

Components – user fields (Z2AG_KOMPL_USRFLD000X000)

User fields offer the possibility to file further customer-specific information in HYDRA besides the fields that are available in the standard of HYDRA. The segment Z2AG_KOMPL_USRFLD000X000 serves for the transfer of these data from the SAP system to HYDRA and for defining them at the series of components.

What user fields are concerned here and which meaning they have is determined via the so-called user field key. Each user field key describes a combination of user fields. The customer-specific configuration of the user field key is carried out in the scope of the HYDRA customizing.

Information

- When it comes to the data exchange of user fields between the SAP system and HYDRA the customer has to guarantee that the user field keys are maintained equally in both systems so that a consistent data exchange is possible between both systems.
- You need at least MLE-Variant ZPPORDER_SAP_007 to support this segment processing

Field	V	T	L	D	Description
AUNR	S	CHAR	12		(Production-)Order number
AFOLG	S	NUMC	6		Sequence
AGNR	S	CHAR	4		Operation
UVGNR	S	CHAR	4		Sub-operation
ATK	S	CHAR	40		Material number; alphabetical characters in UPPER CASE

Field	V	T	L	D	Description
SLP	S	CHAR	10		Bill of material item / Position MPL-RF: Should position of the component in the layer structure (lamination)
USRFLD	S	CHAR	8		User field key
FU_1	K	DATE	8		User field 1
FU_2	K	DATE	8		User field 2
FU_3	K	DATE	8		User field 3
FU_4	K	DATE	8		User field 4
FU_5	K	DATE	8		User field 5
FU_6	K	DATE	8		User field 6
FU_7	K	NUM	8		User field 7
FU_8	K	NUM	8		User field 8
FU_9	K	NUM	8		User field 9
FU_10	K	NUM	8		User field 10
FU_11	K	NUM	8		User field 11
FU_12	K	NUM	8		User field 12
FU_13	K	NUM	8		User field 13
FU_14	K	NUM	8		User field 14
FU_15	K	NUM	8		User field 15
FU_16	K	NUM	8		User field 16
FU_17	K	NUM	8		User field 17
FU_18	K	NUM	8		User field 18
FU_19	K	NUM	8		User field 19
FU_20	K	NUM	8		User field 20
FU_21	K	NUM	8		User field 21
FU_22	K	NUM	8		User field 22
FU_23	K	DEC	13	3	User field 23
FU_24	K	DEC	13	3	User field 24
FU_25	K	DEC	13	3	User field 25
FU_26	K	DEC	13	3	User field 26
FU_27	K	DEC	13	3	User field 27
FU_28	K	DEC	13	3	User field 28
FU_29	K	CHAR	1		User field 29
FU_30	K	CHAR	1		User field 30
FU_31	K	CHAR	1		User field 31
FU_32	K	CHAR	1		User field 32
FU_33	K	CHAR	1		User field 33
FU_34	K	CHAR	1		User field 34
FU_35	K	CHAR	1		User field 35
FU_36	K	CHAR	1		User field 36
FU_37	K	CHAR	1		User field 37
FU_38	K	CHAR	1		User field 38
FU_39	K	CHAR	1		User field 39
FU_40	K	CHAR	1		User field 40
FU_41	K	CHAR	1		User field 41
FU_42	K	CHAR	1		User field 42
FU_43	K	CHAR	1		User field 43
FU_44	K	CHAR	1		User field 44
FU_45	K	CHAR	10		User field 45

Field	V	T	L	D	Description
FU_46	K	CHAR	10		User field 46
FU_47	K	CHAR	10		User field 47
FU_48	K	CHAR	10		User field 48
FU_49	K	CHAR	10		User field 49
FU_50	K	CHAR	10		User field 50
FU_51	K	CHAR	20		User field 51
FU_52	K	CHAR	20		User field 52
FU_53	K	CHAR	20		User field 53
FU_54	K	CHAR	20		User field 54
FU_55	K	CHAR	20		User field 55
FU_56	K	CHAR	20		User field 56
FU_57	K	CHAR	20		User field 57
FU_58	K	CHAR	20		User field 58
FU_59	K	CHAR	20		User field 59
FU_60	K	CHAR	20		User field 60
FU_61	K	CHAR	20		User field 61
FU_62	K	CHAR	20		User field 62
FU_63	K	CHAR	20		User field 63
FU_64	K	CHAR	20		User field 64
FU_65	K	CHAR	40		User field 65
FU_66	K	CHAR	40		User field 66

Production resources / tools (Z2AG_FHM000X000)

Prerequisite for transferring production resources and tools from SAP to HYDRA ist he usage of the HYDRA modules Tools and Resources Management (WRM) or the usage of the module DNC.

Field	V	T	L	D	Description
AUNR	S	CHAR	12		(Production-)Order number
AFOLG	S	NUMC	6		Sequence
AGNR	S	CHAR	4		Operation
UVGNR	S	CHAR	4		Sub-operation
RESTYP	S	CHAR	4		Resource type; possible values: DNC DNC-program ENT Withdrawal unit TEM Temperature control unit VOR Fixture WNR Tool Further resource types (idents) can be defined in the configuration of the resource type (menu <i>WRM: Basic data > Resource types</i>) if the HYDRA tool and resource management (WRM) is used.
ATK	S	CHAR	40		Resource number/material number; alphabetical characters UPPER CASE
ATKBEZ	K	CHAR	40		Designation
BEZ	K	CHAR	30		Comment 1
BEZ_2	K	CHAR	30		Comment 2
SGR_GUT	M	QUAN	13	3	Required quantity

Field	V	T	L	D	Description
SGE_GUT	K	CHAR	3		Quantity unit

Documents (Z2AG_DOC000X000)

Field	V	T	L	D	Description
AUNR	S	CHAR	12		(Production-)Order number
AFOLG	S	NUMC	6		Sequence
AGNR	S	CHAR	4		Operation
UVGNR	S	CHAR	4		Sub-operation
ATK	S	CHAR	40		Document ID: distinct key alphabetical characters UPPER CASE
ATKBEZ	M	CHAR	40		Designation
BEZ	K	CHAR	30		Comment 1
BEZ_2	K	CHAR	30		Comment 2
PATH	M	CHAR	8		Reference to a mapping that is defined in the configuration of the mapping (menu file > <i>System administration</i> > <i>Mappings</i>).
DATEI	M	CHAR	128		File name incl. file extension

User fields of operations (Z2AG_USRFLD000X000)

User fields offer the possibility to file further customer-specific information in HYDRA besides the fields that are available in the standard of HYDRA. The segment Z2AU_USRFLD000X000 serves for the transfer of these data from the SAP system to HYDRA and for defining them at the series of operations.

What user fields are concerned here and which meaning they have is determined via the so-called user field key. Each user field key describes a combination of user fields. The customer-specific configuration of the user field key is carried out in the scope of the HYDRA customizing.

Information

When it comes to the data exchange of user fields between the SAP system and HYDRA the customer has to guarantee that the user field keys are maintained equally in both systems so that a consistent data exchange is possible between both systems.

Field	V	T	L	D	Description
AUNR	S	CHAR	12		(Production-)Order number
AFOLG	S	NUMC	6		Sequence
AGNR	S	CHAR	4		Operation
UVGNR	S	CHAR	4		Sub-operation
USRFLD	S	CHAR	8		User field key
FU_1	K	DATE	8		User field 1

Field	V	T	L	D	Description
FU_2	K	DATE	8		User field 2
FU_3	K	DATE	8		User field 3
FU_4	K	DATE	8		User field 4
FU_5	K	DATE	8		User field 5
FU_6	K	DATE	8		User field 6
FU_7	K	NUM	8		User field 7
FU_8	K	NUM	8		User field 8
FU_9	K	NUM	8		User field 9
FU_10	K	NUM	8		User field 10
FU_11	K	NUM	8		User field 11
FU_12	K	NUM	8		User field 12
FU_13	K	NUM	8		User field 13
FU_14	K	NUM	8		User field 14
FU_15	K	NUM	8		User field 15
FU_16	K	NUM	8		User field 16
FU_17	K	NUM	8		User field 17
FU_18	K	NUM	8		User field 18
FU_19	K	NUM	8		User field 19
FU_20	K	NUM	8		User field 20
FU_21	K	NUM	8		User field 21
FU_22	K	NUM	8		User field 22
FU_23	K	DEC	13	3	User field 23
FU_24	K	DEC	13	3	User field 24
FU_25	K	DEC	13	3	User field 25
FU_26	K	DEC	13	3	User field 26
FU_27	K	DEC	13	3	User field 27
FU_28	K	DEC	13	3	User field 28
FU_29	K	CHAR	1		User field 29
FU_30	K	CHAR	1		User field 30
FU_31	K	CHAR	1		User field 31
FU_32	K	CHAR	1		User field 32
FU_33	K	CHAR	1		User field 33
FU_34	K	CHAR	1		User field 34
FU_35	K	CHAR	1		User field 35
FU_36	K	CHAR	1		User field 36
FU_37	K	CHAR	1		User field 37
FU_38	K	CHAR	1		User field 38
FU_39	K	CHAR	1		User field 39
FU_40	K	CHAR	1		User field 40
FU_41	K	CHAR	1		User field 41
FU_42	K	CHAR	1		User field 42
FU_43	K	CHAR	1		User field 43
FU_44	K	CHAR	1		User field 44
FU_45	K	CHAR	10		User field 45
FU_46	K	CHAR	10		User field 46
FU_47	K	CHAR	10		User field 47
FU_48	K	CHAR	10		User field 48

Field	V	T	L	D	Description
FU_49	K	CHAR	10		User field 49
FU_50	K	CHAR	10		User field 50
FU_51	K	CHAR	20		User field 51
FU_52	K	CHAR	20		User field 52
FU_53	K	CHAR	20		User field 53 This field is displayed in the BDE terminal in the field comment 1
FU_54	K	CHAR	20		User field 54 This field is displayed in the BDE terminal in the field comment 2.
FU_55	K	CHAR	20		User field 55
FU_56	K	CHAR	20		User field 56
FU_57	K	CHAR	20		User field 57
FU_58	K	CHAR	20		User field 58
FU_59	K	CHAR	20		User field 59
FU_60	K	CHAR	20		User field 60
FU_61	K	CHAR	20		User field 61
FU_62	K	CHAR	20		User field 62
FU_63	K	CHAR	20		User field 63
FU_64	K	CHAR	20		User field 64
FU_65	K	CHAR	40		User field 65
FU_66	K	CHAR	40		User field 66

Specific data for the coil-based manufacturing (Z2AG_RF000X000)

This is a segment that depends on the series of operations, i.e. the data transferred here are taken over to operation-specific fields in HYDRA. The data are required if MPL-RF is used.

Field	V	T	L	D	Description
AUNR	S	CHAR	12		(Production-)Order number
AFOLG	S	NUMC	6		Sequence
AGNR	S	CHAR	4		Operation
UVGNR	S	CHAR	4		Sub-operation
RFA GTYP	K	CHAR	1		Flag → type of operation: " " No special processing "P" Packaging operation
RFABZ	MM	CHAR	1		Distinguishes mother and children OPs in case of a planned deduction. "M" Mother OP of a planned deduction "K" Child OP of a planned deduction (in this case a special material movement - 531 - is processed).
RFOPT_RS	MM	CHAR	1		Indicator for cutting coils (only relevant if it is a cutting OP) " " no cutting "T" cutting of coils active (numbering of daughter coil) "M" cutting active (mother coils are generated again)
RFMANR	MM	CHAR	40		Cutting plan (order combination): The deduction (link) is assigned to the respective mother OP by means of this field and the following fields in case of a planned deduction. The mother OP references itself.
RFTRANZ	MM	NUMC	5		In case of cutting operations: number of the planned daughter coils per cut.

Field	V	T	L	D	Description
RFTRANZSUM	MM	NUMC	5		In case of cutting operations (mother OP): number of the planned daughter coils per cut (beyond all deductions/ operations).
RFRANZ	MM	NUMC	6		Planned number of coils (only information)
RFSTKF	MM	NUMC	8		Area of a piece Unit: MM2 / ST (integer)
RFBSBRS	MM	DEC	10	3	Sum seam width (Sum "border width" If several coils are produced simultaneously in one series of operations this field contains the sum of the single seam widths. The seam width of the individual series of operations is explicitly set for each series of operations ("mother" and "child" series of operations) in case of deductions (no totals formation) Unit: MM
RFBREITEE	MM	DEC	10	3	Input-width operation Unit: MM
RFBREITEA	MM	DEC	10	3	Output-width operation Width of one outgoing roll within the operation (even if more than one roll is cut). The initial width of the individual operations is explicitly set for <i>each</i> operation ("mother" and "child" operations) in case of planned deductions (no totals formation). Unit: MM
RFAGVFA	MM	DEC	10	3	Mass per unit area Unit: G/M2
Hülsengewicht	K	DEC	10	3	The weight of the casing of the daughter rolls is specified here in case of cutting processes

Order network (Z2AG_ANETZ000X000)

This segment is an order-wide segment. It can be used to define order-wide relationships. These will be accounted for in the planning in the HYDRA shop floor scheduling system (HLS) and/or in processing (e.g. target quantity update, if active) when the involved operations are planned (ade_verarb_code.planen <> "N") and/or entered (auftrags_bestand.erfassbar = "J"). The transfer of the data is only possible if the transferred orders and/or their operations exist already in HYDRA.



You do not have to transfer order network relationships between neighboring operations within one production order, since the system creates these relationships automatically.

Feld	V	T	L	D	Beschreibung
SAPAUNRV	S	CHAR	12		Order number of the predecessor OP
SAPAFOLGV	S	CHAR	6		Sequence number of the predecessor OP
SAPVGNRV	S	CHAR	4		Operation number of the predecessor OP
SAPUVGNRV	S	CHAR	4		Sub operation number of the predecessor OP
SAPAUNRN	S	CHAR	12		order number of the successor OP
SAPAFOLGN	S	CHAR	6		Sequence number of the successor OP
SAPVGNRN	S	CHAR	4		Operation number of the successor OP

Feld	V	T	L	D	Beschreibung
SAPUVGNRN	S	CHAR	4		Sub operation number of the successor OP
AOB	K	CHAR	2		Fix „ES“

Production variants (Z2MD_PRODVAR000X000)

This segment is used to enter / to maintain production variants in HYDRA. If configured accordingly , these variants will be accounted for in the determination of production variants when orders are transferred.

Field	V	T	L	D	Description
VER	S	CHAR	10		Version
STA	M	CHAR	1		Status of the production variant F = Released S = Blocked
ATK	S	CHAR	40		Article
MATTYP	S	CHAR	10		Material type; blank (currently not processed)
MNR	S	CHAR	8		Machine
MGRP	S	CHAR	8		Group
MANZ	K	DEC	13	3	Number of machines; "1" (currently not processed)
RES	M	CHAR	40		Resource. Please note: The default resource will always be occupied with the resource type WNR.
RESFAM	K	CHAR	18		Resource family (currently not processed)
WANZ	K	NUM	8		Number of resources
SZY	K	NUM	8		Target cycle in [seconds/ 1000 cycles]
TLG	K	DEC	13	3	Partitioning
RUEZ	K	NUM	8		Setup time (seconds)
ABRZ	K	NUM	8		Dismantling/retooling time (seconds)
DSBEZ	SA	CHAR	15		Data identifier
BEM	K	CHAR	50		Comment
PRI0	M	NUMC	1		Priority
DATB	K	DATE	8		Valid from
DATE	K	DATE	8		Valid to

Deleting operations (E2BP_PP_PDC_OPERA1000)

The structure described in the following, controls the deletion process for already transferred production orders and/or their operations in the subsystem.

Field name	T	L	Description	Usage in HYDRA
SOURCE_SYS	CHAR	10	Logical system	Not used
CONF_NO	NUMC	10	Confirmation number of the operation	Confirmation no.

Field name	T	L	Description	Usage in HYDRA
ORDERID	CHAR	12	Order	According to configuration (*1)
SEQUENCE	CHAR	6	Sequence	According to configuration (*1)
OPERATION	CHAR	4	Operation	According to configuration (*1)
SUB_OPER	CHAR	4	Sub-operation	According to configuration (*1)
SUBSYSTEM_GROUPING	CHAR	3	Grouping subsystem connection	Restriction to a subsystem group possible.